# Companies Data Management System

## Python + Pandas Based Data Analysis Project

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## Project Overview

This project is a beginner-friendly Python data analysis system built using Pandas and Matplotlib. The system allows users to manage and analyze company-related data stored in CSV format.

The project demonstrates: CSV file handling Data analysis using Pandas Data visualization using Matplotlib Data manipulation and sorting Menu-driven Python programming

## Main Features

| Feature | Description |
|---|---|
| Read CSV Data | Displays complete company dataset |
| Top/Bottom Records | Shows selected rows from dataset |
| Sorting | Sorts records alphabetically |
| Line Chart | Visualizes company data using line graph |
| Bar Chart | Visualizes company data using bar graph |
| Update Data | Allows adding new column data |
| Delete Column | Removes selected column |

## Tech Stack

**Programming Language:** Python
**Libraries Used:** Pandas, NumPy, Matplotlib
**Data Storage:** CSV File
**IDE:** VS Code / Any Python IDE

# EXCEL VIEW

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Code | Company Name | Employee Number | Net worth in billion dollars | Global Position |
| 2 | 112 | Microsoft | 221000 | 2135 | 21 |
| 3 | 113 | Meta | 58604 | 36 | 27 |
| 4 | 114 | Amazon | 1468000 | 1053.5 | 1 |
| 5 | 115 | Google | 150028 | 1420 | 2 |
| 6 | 116 | Tesla | 110000 | 710.78 | 103 |
| 7 | 117 | Tata | 935000 | 260 | 78 |
| 8 | 118 | Reliance | 342982 | 229 | 104 |
| 9 | 119 | Infosys | 276319 | 79.06 | 50 |
| 10 | 120 | Spacex | 12000 | 2 | 387 |
| 11 | 121 | Netflix | 11300 | 131.6 | 245 |

Companies Data Management

# CSV VIEW

```
Code,Company Name,Employee Number,Net worth in billion
dollars,Global Position
112,Microsoft,221000,2135,21
113,Meta,58604,36,27
114,Amazon,1468000,1053.5,1
115,Google,150028,1420,2
116,Tesla,110000,710.78,103
117,Tata,935000,260,78
118,Reliance,342982,229,104
119,Infosys,276319,79.06,50
120,Spacex,12000,2,387
121,Netflix,11300,131.6,245
```

Companies Data Management.CSV

# CODING

```python
PROJECT.PY > no_index
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def main_menu():
    print("--------------")
    print("Companies Data")
    print("--------------""\n")
    print("Data Analysis")
    print("1.Reading complete csv report")
    print("2.Display Record of top MNC's without index")
    print("-----------""\n")
    print("Data Visualization")
    print("3.Line chart")
    print("4.Bar chart")
    print("----- --""\n")
    print("Data Manipulation")
    print("5.Specific number of companies from the top")
    print("6.Specific number of companies form bottom")
    print("7.Sorting the data as per your choice")
    print("8.Updating_Adding_new_data")
    print("9.Deleting_the_column")
    print("10.specific_colummn")

main_menu()

def ReadCSV():
    print("Reading complete csv report")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    print(df)


def no_index():
    print("Display record of top MNC's")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv",index_col=0)
    print(df)
```

```python
def line_chart():
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    x=['Microsoft','Meta','Amazon','Google','Tesla','Tata','Reliance','Infosys','SpaceX','Netflix']
    y=[221000,58604,1468000,150028,110000,935000,342982,276319,12000,11300]
    plt.plot(x,y)
    plt.xlabel('Company Name')
    plt.ylabel('No. of employs in Lakh')
    plt.title("No. of employees in top MNCs")
    plt.show()


def Bar_chart():
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    x=['Microsoft','Meta','Amazon','Google','Tesla','Tata','Reliance','Infosys','SpaceX','Netflix']
    y=[2135,36,1053.5,1420,710.78,260,229,79.06,2,131.6]
    plt.bar(x,y)
    plt.xlabel("COMPANY NAME")
    plt.ylabel("NET WORTH IN BILLION DOLLARS")
    plt.title("Net worth of top MNCs")
    plt.show()


def companies_from_top():
    print("Specific number of companies from the top")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    n=int(input("Enter how many companies you want to display from the top:"))
    print(df.head(n))


def companies_from_bottom():
    print("Specific number of companies form bottom")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    n=int(input("Enter how many companies you want to display from the bottom:"))
    print(df.tail(n))


```

```python
def data_sorting():
    print("Record as per company name")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    df.sort_values(["Company Name"],inplace=True)
    print(df,"\n","\n")

    print("Record as per employee Number")
    df.sort_values(["Employee Number"],inplace=True)
    print(df,"\n","\n")

    print("Record as per Net Worth in Billions Dollors")
    df.sort_values(["Net worth in billion dollars"],inplace=True)
    print(df,"\n","\n")

    print("Record as per Global Position")
    df.sort_values(["Global Position"],inplace=True)
    print(df,"\n","\n")


def Updating_Adding_new_data():
    print("updating or adding new data")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    print("Enter details")
    h=input("Enter column name:")
    det=eval(input("Enter details corresponding to all subjects : (enclosed in []:"))
    df[h]=pd.Series(data=det,index=df.index)
    print(df)
    print("column inserted")


```

```python
def Deleting_the_column():
    print("Deleting the column")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    a=input("Enter column name which need to be deleted: ")
    df.drop([a],axis=1,inplace=True)
    print(df)
    print("column deleted")

```

```python
def specific_company():
    print("Search for details of a specific company")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    col=input("Enter ccompany name whose detail you want to see:")
    print(df[col])
```

```python
def Column_Renaming():
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    print(df)
    n=input("Enter old column name: ")
    N=input("Enter new column name: ")
    df.rename(columns = {n:N}, inplace = True)
    print("\nAfter modifying column:\n",)
    print(df)
    print("Column renamed")
```

```python
def Deleting_the_row():
    print("Deleting the row")
    df=pd.read_csv(r"C:\Users\Panshul Sharma\Documents\CSVCompanies Data.csv")
    a=input("Enter Company Name which need to be deleted: ")
    df.drop(df.index[(df["Company Name"] == "a")],axis=0,inplace=True)
    print(df)
    print("column deleted")
```

```python
option=True
while option:
    opt=int(input("Enter your choice "))
ject File\Class 12\paper.py 1:
        ReadCSV()
    elif opt==2:
        no_index()
    elif opt==3:
        line_chart()
    elif opt==4:
        Bar_chart()
    elif opt==5:
        companies_from_top()
    elif opt==6:
        companies_from_bottom()
    elif opt==7:
        data_sorting()
    elif opt==8:
        Updating_Adding_new_data()
    elif opt==9:
        Deleting_the_column()
    elif opt==10:
        specific_colummn()
    else:
        opt!=opt
        print("INVALID NUMBER!!!")

    abc=input("type y to continue or n to exit ")

    if abc=="y":
        option=True

    else:

        option=False
```

# OUTPUT

```
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Companies Data
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Data Analysis
1.Reading complete csv report
2.Display Record of top MNC's without index
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Data Visualization
3.Line chart
4.Bar chart
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Data Manipulation
5.Specific number of companies from the top
6.Specific number of companies form bottom
7.Sorting the data as per your choice
8.Updating_Adding_new_data
9.Deleting_the_column
10.specific_colummn
~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~


Enter your choice
```

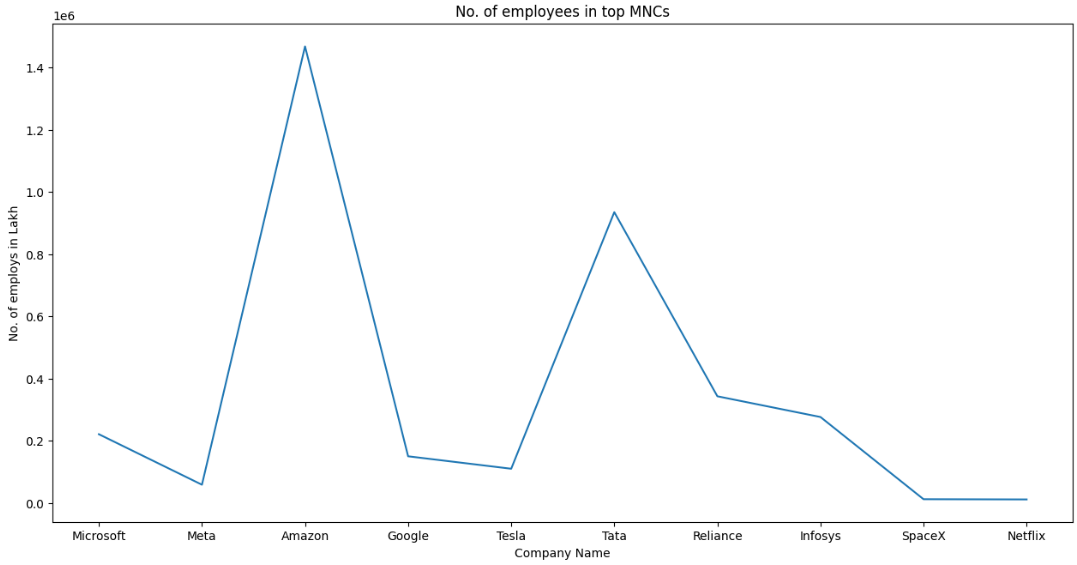
```
Enter your choice 1
Reading complete csv report
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
0   112    Microsoft           221000                       2135.00               21
1   113         Meta            58604                         36.00               27
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
4   116        Tesla           110000                        710.78              103
5   117         Tata           935000                        260.00               78
6   118     Reliance           342982                        229.00              104
7   119      Infosys           276319                         79.06               50
8   120       Spacex            12000                          2.00              387
9   121      Netflix            11300                        131.60              245
type y to continue or n to exit
```

```
type y to continue or n to exit y
Enter your choice 2
Display record of top MNC's
     Company Name  Employee Number  Net worth in billion dollars  Global Position
Code
112     Microsoft           221000                       2135.00               21
113          Meta            58604                         36.00               27
114        Amazon          1468000                       1053.50                1
115        Google           150028                       1420.00                2
116         Tesla           110000                        710.78              103
117          Tata           935000                        260.00               78
118      Reliance           342982                        229.00              104
119       Infosys           276319                         79.06               50
120        Spacex            12000                          2.00              387
121       Netflix            11300                        131.60              245
type y to continue or n to exit
```

```
Enter your choice 3
type y to continue or n to exit y
```

# Choice:3

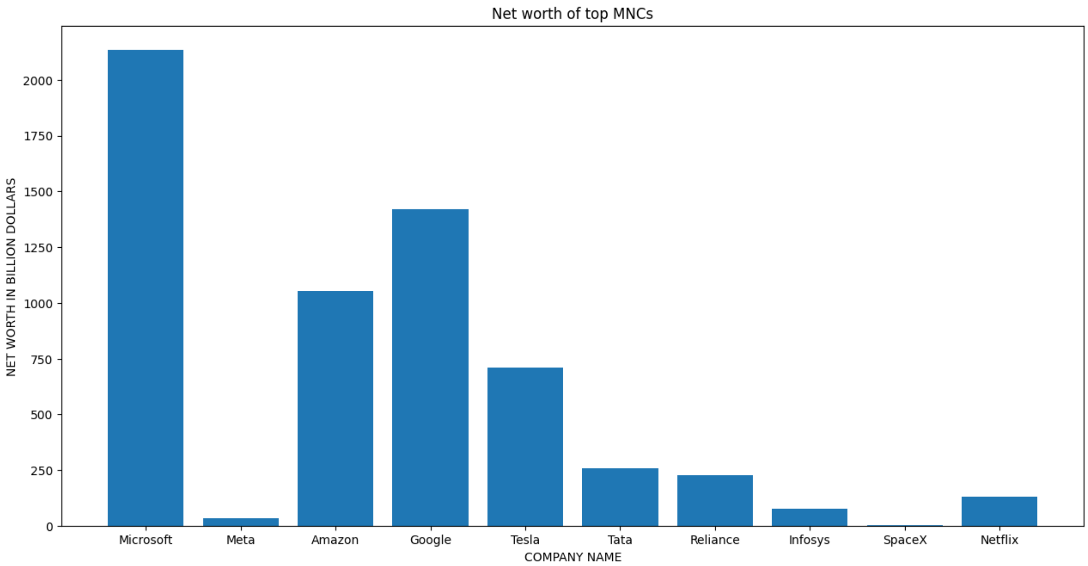
# Line chart




```
Enter your choice 4
type y to continue or n to exit
```

# Choice:4

# Bar Chart



```
Enter your choice 5
Specific number of companies from the top
Enter how many companies you want to display from the top:5
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
0   112    Microsoft           221000                       2135.00               21
1   113         Meta            58604                         36.00               27
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
4   116        Tesla           110000                        710.78              103
type y to continue or n to exit
```

```
Enter your choice 6
Specific number of companies form bottom
Enter how many companies you want to display from the bottom:5
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
5   117         Tata           935000                        260.00               78
6   118     Reliance           342982                        229.00              104
7   119      Infosys           276319                         79.06               50
8   120       Spacex            12000                          2.00              387
9   121      Netflix            11300                        131.60              245
type y to continue or n to exit
```

```
Enter your choice 7
Record as per company name
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
7   119      Infosys           276319                         79.06               50
1   113         Meta            58604                         36.00               27
0   112    Microsoft           221000                       2135.00               21
9   121      Netflix            11300                        131.60              245
6   118     Reliance           342982                        229.00              104
8   120       Spacex            12000                          2.00              387
5   117         Tata           935000                        260.00               78
4   116        Tesla           110000                        710.78              103


Record as per employee Number
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
9   121      Netflix            11300                        131.60              245
8   120       Spacex            12000                          2.00              387
1   113         Meta            58604                         36.00               27
4   116        Tesla           110000                        710.78              103
3   115       Google           150028                       1420.00                2
0   112    Microsoft           221000                       2135.00               21
7   119      Infosys           276319                         79.06               50
6   118     Reliance           342982                        229.00              104
5   117         Tata           935000                        260.00               78
2   114       Amazon          1468000                       1053.50                1


Record as per Net Worth in Billions Dollors
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
8   120       Spacex            12000                          2.00              387
1   113         Meta            58604                         36.00               27
7   119      Infosys           276319                         79.06               50
9   121      Netflix            11300                        131.60              245
6   118     Reliance           342982                        229.00              104
5   117         Tata           935000                        260.00               78
4   116        Tesla           110000                        710.78              103
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
0   112    Microsoft           221000                       2135.00               21
```

```
Record as per Global Position
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
0   112    Microsoft           221000                       2135.00               21
1   113         Meta            58604                         36.00               27
7   119      Infosys           276319                         79.06               50
5   117         Tata           935000                        260.00               78
4   116        Tesla           110000                        710.78              103
6   118     Reliance           342982                        229.00              104
9   121      Netflix            11300                        131.60              245
8   120       Spacex            12000                          2.00              387


type y to continue or n to exit
```

```
Enter your choice 8
updating or adding new data
Enter details
Enter column name:Year founded in
Enter details corresponding to all subjects : (enclosed in []:[1994,1998,1975,2004,1981,1868,2003,1958,1997,2002]
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position  Year founded in
0   112    Microsoft           221000                       2135.00               21             1994
1   113         Meta            58604                         36.00               27             1998
2   114       Amazon          1468000                       1053.50                1             1975
3   115       Google           150028                       1420.00                2             2004
4   116        Tesla           110000                        710.78              103             1981
5   117         Tata           935000                        260.00               78             1868
6   118     Reliance           342982                        229.00              104             2003
7   119      Infosys           276319                         79.06               50             1958
8   120       Spacex            12000                          2.00              387             1997
9   121      Netflix            11300                        131.60              245             2002
column inserted
type y to continue or n to exit
```

```
Enter your choice 9
Deleting the column
Enter column name which need to be deleted: Code
  Company Name  Employee Number  Net worth in billion dollars  Global Position
0    Microsoft           221000                       2135.00               21
1         Meta            58604                         36.00               27
2       Amazon          1468000                       1053.50                1
3       Google           150028                       1420.00                2
4        Tesla           110000                        710.78              103
5         Tata           935000                        260.00               78
6     Reliance           342982                        229.00              104
7      Infosys           276319                         79.06               50
8       Spacex            12000                          2.00              387
9      Netflix            11300                        131.60              245
column deleted
type y to continue or n to exit
```

```
Enter your choice 10
Read with specific column
Enter column whose detail you want to see:Company Name
0    Microsoft
1         Meta
2       Amazon
3       Google
4        Tesla
5         Tata
6     Reliance
7      Infosys
8       Spacex
9      Netflix
Name: Company Name, dtype: object
type y to continue or n to exit
```

```
Enter your choice 11
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
0   112    Microsoft           221000                       2135.00               21
1   113         Meta            58604                         36.00               27
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
4   116        Tesla           110000                        710.78              103
5   117         Tata           935000                        260.00               78
6   118     Reliance           342982                        229.00              104
7   119      Infosys           276319                         79.06               50
8   120       Spacex            12000                          2.00              387
9   121      Netflix            11300                        131.60              245
Enter old column name: Code
Enter new column name: S.no

After modifying column:

   S.no Company Name  Employee Number  Net worth in billion dollars  Global Position
0   112    Microsoft           221000                       2135.00               21
1   113         Meta            58604                         36.00               27
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
4   116        Tesla           110000                        710.78              103
5   117         Tata           935000                        260.00               78
6   118     Reliance           342982                        229.00              104
7   119      Infosys           276319                         79.06               50
8   120       Spacex            12000                          2.00              387
9   121      Netflix            11300                        131.60              245
Column renamed
type y to continue or n to exit y
```

```
Enter your choice 12
Deleting the row
Enter Company Name which need to be deleted: Tesla
   Code Company Name  Employee Number  Net worth in billion dollars  Global Position
0   112    Microsoft           221000                       2135.00               21
1   113         Meta            58604                         36.00               27
2   114       Amazon          1468000                       1053.50                1
3   115       Google           150028                       1420.00                2
5   117         Tata           935000                        260.00               78
6   118     Reliance           342982                        229.00              104
7   119      Infosys           276319                         79.06               50
8   120       Spacex            12000                          2.00              387
9   121      Netflix            11300                        131.60              245
column deleted
type y to continue or n to exit n
PS C:\Python Project File\Class 12>
```